

Problem 3

We are given the discrete probability distribution for the sequence $(X_k)_{k \in \mathbb{N}}$:

$$\mathbb{P}(X_k = k^\lambda) = \mathbb{P}(X_k = -k^\lambda) = \frac{1}{2}. \quad (1)$$

Since these probabilities sum to 1, X_k is a discrete random variable taking only the values k^λ and $-k^\lambda$. Consequently, its absolute value is deterministic $\mathbb{P}$ -almost surely:

$$|X_k| = k^\lambda \quad \mathbb{P}\text{-a.s.} \quad (2)$$

Part a)

We verify the four hypotheses of the Strong Law of Large Numbers (Theorem 4.3.5):

- (i) $X_k \in L^2(\Omega, \mathcal{F}, \mathbb{P}) \forall k \in \mathbb{N}, \lambda \leq 0$. Since $k \geq 1$ and $\lambda \leq 0$, we have that $k^\lambda \leq 1$. Combining this with (2), we get $|X_k| \leq 1$ $\mathbb{P}$ -almost surely, and, therefore, the second moment is bounded:

$$\mathbb{E}(X_k^2) = k^{2\lambda} \leq 1 < \infty, \quad (3)$$

i.e., $X_k \in L^2(\Omega, \mathcal{F}, \mathbb{P}) \forall k \in \mathbb{N}$.

- (ii) **The expected values all equal zero** ($\lambda \in \mathbb{R}, \forall k \in \mathbb{N}$). Using the discrete distribution defined in (1), we apply the definition of the expected value (Definition 3.3.1):

$$\mathbb{E}(X_k) = k^\lambda \cdot \mathbb{P}(X_k = k^\lambda) + (-k^\lambda) \cdot \mathbb{P}(X_k = -k^\lambda) = k^\lambda \left(\frac{1}{2}\right) - k^\lambda \left(\frac{1}{2}\right) = 0 \quad (4)$$

for every $k \in \mathbb{N}$. In particular, all X_k share the same expected value $\mu := 0$. Note that this holds for any $\lambda \in \mathbb{R}$, not just $\lambda \leq 0$.

- (iii) **The variances are uniformly bounded** ($\lambda \leq 0, \forall k \in \mathbb{N}$). Using (3) and (4), the variance of X_k is:

$$\text{Var}(X_k) = \mathbb{E}(X_k^2) - (\mathbb{E}(X_k))^2 = k^{2\lambda} - 0 = k^{2\lambda}.$$

As established in Part (i), since $k \geq 1$ and $\lambda \leq 0$, the same exponential bounding argument applied to $2\lambda \leq 0$ guarantees that $k^{2\lambda} \leq 1$ for every $k \in \mathbb{N}$. Therefore, the sequence of variances is uniformly bounded by a constant c :

$$c := \sup_{k \in \mathbb{N}} \text{Var}(X_k) = \sup_{k \in \mathbb{N}} k^{2\lambda} \leq 1 < \infty.$$

- (iv) **The X_k are pairwise uncorrelated** ($\lambda \in \mathbb{R}, \forall k \in \mathbb{N}$). The random variables $(X_k)_{k \in \mathbb{N}}$ are given to be independent. Since we established $X_k \in L^2(\Omega, \mathcal{F}, \mathbb{P})$ in (i), and $L^2(\Omega, \mathcal{F}, \mathbb{P}) \subset L^1(\Omega, \mathcal{F}, \mathbb{P})$ (Corollary 3.3.20), they are uncorrelated (Corollary 3.3.33).

Since all hypotheses are satisfied, Theorem 4.3.5 therefore gives:

$$M_n := \frac{1}{n} \sum_{k=1}^n X_k \xrightarrow{\mathbb{P}\text{-a.s.}} \mu = 0.$$

□

Part b)

Convergence in Probability

We want to show that for all $\varepsilon > 0$, $M_n \xrightarrow{p} 0$, i.e. (Definition 4.1.1.) for $\lambda < 1/2$:

$$\mathbb{P}(|M_n| \geq \varepsilon) \xrightarrow{n \rightarrow \infty} 0.$$

For the variance of M_n , we have:

$$\begin{aligned} \text{Var}(M_n) &= \text{Var}\left(\frac{1}{n} \sum_{k=1}^n X_k\right) \\ &= \frac{1}{n^2} \text{Var}\left(\sum_{k=1}^n X_k\right) \quad (\text{by Remark 3.3.16}) \\ &= \frac{1}{n^2} \sum_{k=1}^n \text{Var}(X_k) \quad (\text{by Theorem 3.3.35 (ii) since } X_k \text{ are independent}) \\ &= \frac{1}{n^2} \sum_{k=1}^n k^{2\lambda} \quad (\text{substituting } \text{Var}(X_k) = k^{2\lambda} \text{ from part a (iii)}) \end{aligned} \quad (5)$$

We have two cases:

- **Case $\lambda \leq 0$:** In part a) we show that $M_n \xrightarrow{\mathbb{P}\text{-a.s.}} 0$ for $\lambda \leq 0$. Therefore, we have $M_n \xrightarrow{p} 0$ for $\lambda \leq 0$ (Theorem 4.1.2a).
- **Case $0 < \lambda < \frac{1}{2}$:** Since $\lambda > 0$, the exponent $2\lambda > 0$, making the map $x \mapsto x^{2\lambda}$ strictly monotonically increasing on $[1, \infty)$. We bound the sum from above using an integral comparison:

$$\sum_{k=1}^n k^{2\lambda} \leq \int_1^{n+1} x^{2\lambda} dx = \frac{(n+1)^{2\lambda+1} - 1}{2\lambda + 1} \leq \frac{(n+1)^{2\lambda+1}}{2\lambda + 1}.$$

Combining this with (5) yields:

$$\text{Var}(M_n) \leq \frac{(n+1)^{2\lambda+1}}{(2\lambda + 1) n^2} \xrightarrow{n \rightarrow \infty} 0,$$

because $\lambda < 1/2$ implies $2\lambda + 1 < 2$, meaning the denominator n^2 grows strictly faster than the numerator. Since $\text{Var}(M_n) \geq 0$, we have:

$$\text{Var}(M_n) \xrightarrow{n \rightarrow \infty} 0 \quad (6)$$

From part a (ii), we have $\mathbb{E}(X_k) = 0$ for every $k \in \mathbb{N}$, so $\mathbb{E}(M_n) = 0$. Since $M_n \in L^2(\Omega, \mathcal{F}, \mathbb{P})$ (from part a (i)), we have that for any $\varepsilon > 0$ (Chebyshev's inequality, Theorem 3.3.25):

$$\mathbb{P}(|M_n| \geq \varepsilon) \leq \frac{\text{Var}(M_n)}{\varepsilon^2}$$

Since $\text{Var}(M_n) \xrightarrow{n \rightarrow \infty} 0$ (6), then:

$$\frac{\text{Var}(M_n)}{\varepsilon^2} \xrightarrow{n \rightarrow \infty} 0$$

Since probabilities are non-negative, we have:

$$\begin{aligned} 0 &\leq \mathbb{P}(|M_n| \geq \varepsilon) \leq \frac{\text{Var}(M_n)}{\varepsilon^2} \xrightarrow{n \rightarrow \infty} 0 \\ \implies \mathbb{P}(|M_n| \geq \varepsilon) &\xrightarrow{n \rightarrow \infty} 0. \end{aligned}$$

By Definition 4.1.1, $M_n \xrightarrow{p} 0$.

In both cases, we have shown that $M_n \xrightarrow{p} 0$ for all $\lambda < \frac{1}{2}$. □

Almost Sure Convergence

Yes, $M_n \xrightarrow{\mathbb{P}\text{-a.s.}} 0$ for all $\lambda < \frac{1}{2}$. From part a), we know that $M_n \xrightarrow{\mathbb{P}\text{-a.s.}} 0$ for $\lambda \leq 0$. We now prove the case for $0 < \lambda < \frac{1}{2}$. First, consider the unscaled sum $S_n = \sum_{k=1}^n X_k$ and a positive integer p . We want to investigate the $2p$ -th moment of S_n . Using the multinomial theorem to algebraically expand the $2p$ -th power gives:

$$S_n^{2p} = \left(\sum_{k=1}^n X_k \right)^{2p} = \sum_{a_1 + \dots + a_n = 2p} \frac{(2p)!}{a_1! \dots a_n!} \prod_{k=1}^n X_k^{a_k},$$

where the sum runs over all tuples of non-negative integers $(a_1, \dots, a_n)$ that sum to $2p$. Taking the expected value of both sides, we get:

$$\begin{aligned} \mathbb{E}[S_n^{2p}] &= \mathbb{E} \left[\sum_{a_1 + \dots + a_n = 2p} \frac{(2p)!}{a_1! \dots a_n!} \prod_{k=1}^n X_k^{a_k} \right] \\ \implies \mathbb{E}[S_n^{2p}] &= \sum_{a_1 + \dots + a_n = 2p} \frac{(2p)!}{a_1! \dots a_n!} \mathbb{E} \left[\prod_{k=1}^n X_k^{a_k} \right] \quad (\text{by linearity of expectation}) \\ \implies \mathbb{E}[S_n^{2p}] &= \sum_{a_1 + \dots + a_n = 2p} \frac{(2p)!}{a_1! \dots a_n!} \prod_{k=1}^n \mathbb{E}[X_k^{a_k}] \quad (\text{by independence of } X_k) \end{aligned} \quad (7)$$

Let A be the set of tuples $(a_1, \dots, a_n)$ summing to $2p$ where all a_k are even integers, and let A^c be its complement (tuples where at least one a_k is odd). Because the distribution of X_k in (1) is symmetric around zero, any odd moment is zero (i.e., $\mathbb{E}[X_k^{a_k}] = 0$ if a_k is odd). Consequently, for any tuple in A^c , the entire product in (7) becomes zero. We can therefore restrict the sum in (7) strictly to the set A :

$$\mathbb{E}[S_n^{2p}] = \sum_{(a_1, \dots, a_n) \in A} \frac{(2p)!}{a_1! \dots a_n!} \prod_{k=1}^n \mathbb{E}[X_k^{a_k}]. \quad (8)$$

Next, we establish bounds for the components of this sum.

- (i) **Bounding the product of expected values.** For any $k \leq n$, since $\lambda > 0$, it holds that $k^\lambda \leq n^\lambda$. Using the deterministic bound from (2), we have $|X_k| = k^\lambda \leq n^\lambda$. Therefore:

$$\prod_{k=1}^n \mathbb{E}[X_k^{a_k}] \leq \prod_{k=1}^n \mathbb{E}[(n^\lambda)^{a_k}] = \prod_{k=1}^n n^{\lambda a_k} = n^{\lambda \sum_{k=1}^n a_k} = n^{2p\lambda}.$$

- (ii) **Bounding the number of terms and coefficients.** Since every a_k in A is an even integer, any non-zero a_k must satisfy $a_k \geq 2$. Because the elements sum to $2p$, there can be at most p strictly positive elements in any tuple. If exactly $j \leq p$ variables have strictly positive powers, the number of ways to choose these j indices from the n available variables is $\binom{n}{j}$. Then:

$$\binom{n}{j} = \frac{n!}{j!(n-j)!} = \frac{n(n-1) \dots (n-(j-1)) \cdot (n-j)!}{j!(n-j)!} = \frac{n(n-1) \dots (n-(j-1))}{j!}$$

Since each of the j factors in the numerator is less than or equal to n , the numerator is bounded by n^j . Because $j! \geq 1$ and $j \leq p$, we have:

$$\binom{n}{j} \leq \frac{n^j}{j!} \leq n^j \leq n^p.$$

- (iii) **Bounding the multinomial coefficient.** The multinomial coefficient is bounded by $\frac{(2p)!}{a_1! \dots a_n!} \leq (2p)!$. The total number of ways to partition the integer $2p$ into j positive even integers depends only on p . Thus, we can group all these p -dependent combinatorial factors into a single constant $C_p > 0$ that does not depend on n .

Applying these bounds to (8) yields:

$$\mathbb{E} [S_n^{2p}] \leq C_p \cdot n^p \cdot n^{2p\lambda} = C_p n^{p(2\lambda+1)}.$$

By definition, the sample mean is $M_n = \frac{1}{n} S_n$. Raising both sides to the $2p$ -th power gives $M_n^{2p} = \frac{1}{n^{2p}} S_n^{2p}$. Therefore, the $2p$ -th moment of the sample mean is bounded by:

$$\mathbb{E} [M_n^{2p}] = \frac{1}{n^{2p}} \mathbb{E} [S_n^{2p}] \leq \frac{C_p n^{p(2\lambda+1)}}{n^{2p}} = C_p n^{p(2\lambda-1)}.$$

Now, we apply Markov's Inequality (Theorem 3.3.24) to the non-negative random variable $|M_n|^{2p}$. For any $\varepsilon > 0$:

$$\mathbb{P}(|M_n| \geq \varepsilon) = \mathbb{P}(|M_n|^{2p} \geq \varepsilon^{2p}) \leq \frac{\mathbb{E}(M_n^{2p})}{\varepsilon^{2p}} \leq \frac{C_p}{\varepsilon^{2p}} n^{p(2\lambda-1)}.$$

Since $\lambda < 1/2$, it strictly holds that $2\lambda - 1 < 0$. We can therefore choose an integer p large enough such that the exponent $p(2\lambda - 1) < -1$. This ensures that the probabilities over n form a convergent series (a p -series with exponent strictly less than -1):

$$\sum_{n=1}^{\infty} \mathbb{P}(|M_n| \geq \varepsilon) \leq \frac{C_p}{\varepsilon^{2p}} \sum_{n=1}^{\infty} n^{p(2\lambda-1)} < \infty.$$

Because the sum of probabilities is finite, by the 1st Borel-Cantelli Lemma (Lemma 1.1.17), we have:

$$\mathbb{P} \left(\limsup_{n \rightarrow \infty} \{|M_n| \geq \varepsilon\} \right) = 0.$$

This means that for any fixed $\varepsilon > 0$, with probability 1, the event $\{|M_n| \geq \varepsilon\}$ occurs only finitely often. By taking a countable sequence $\varepsilon_m = 1/m$ for $m \in \mathbb{N}$, the countable union of these null sets is still a null set. Thus, with probability 1, $|M_n|$ eventually remains below $1/m$ for every $m \in \mathbb{N}$, which establishes that M_n converges to zero almost surely: $M_n \xrightarrow{\mathbb{P}\text{-a.s.}} 0$. □

Part c)

Convergence in Distribution.

Step 1: Characteristic function of X_k for $\lambda = \frac{1}{2}$. Setting $\lambda = \frac{1}{2}$, we have $\mathbb{E}(X_k) = 0$ (by (4)) and

$$\mathbb{E}(X_k^2) = k^{2\lambda} \big|_{\lambda=1/2} = k.$$

Since X_k takes only the values $\sqrt{k}$ and $-\sqrt{k}$, each with probability $\frac{1}{2}$, the characteristic function of X_k is (Definition 3.5.1):

$$\begin{aligned} \varphi_{X_k}(t) &= E(e^{itX_k}) = \frac{1}{2} e^{it\sqrt{k}} + \frac{1}{2} e^{-it\sqrt{k}} \\ &= \frac{1}{2} (\cos(t\sqrt{k}) + i \sin(t\sqrt{k})) + \frac{1}{2} (\cos(t\sqrt{k}) - i \sin(t\sqrt{k})) \\ &= \cos(t\sqrt{k}) \quad t \in \mathbb{R}, \end{aligned} \tag{9}$$

Step 2: Characteristic function of M_n . Since $X_1, \dots, X_n$ are independent and $M_n = \frac{1}{n} \sum_{k=1}^n X_k$, we compute, for $u \in \mathbb{R}$, the characteristic function of M_n :

$$\begin{aligned} \varphi_{M_n}(u) &\stackrel{(3.5.1)}{=} E\left(e^{iuM_n}\right) = E\left(e^{i\frac{u}{n} \sum_{k=1}^n X_k}\right) = E\left(\prod_{k=1}^n e^{i\frac{u}{n} X_k}\right) \stackrel{(\text{ind.})}{=} \prod_{k=1}^n E\left(e^{i\frac{u}{n} X_k}\right) \\ &\stackrel{(3.5.1)}{=} \prod_{k=1}^n \varphi_{\frac{1}{n} X_k}(u) \stackrel{(3.5.2b)}{=} \prod_{k=1}^n \varphi_{X_k}\left(\frac{u}{n}\right) \stackrel{(9)}{=} \prod_{k=1}^n \cos\left(\frac{u\sqrt{k}}{n}\right) \end{aligned} \quad (10)$$

Step 3: Computing $\lim_{n \rightarrow \infty} \varphi_{M_n}(u)$. Fix $u \in \mathbb{R}$. For every $k \leq n$,

$$\left|\frac{u\sqrt{k}}{n}\right| = \frac{|u|\sqrt{k}}{n} \leq \frac{|u|\sqrt{n}}{n} = \frac{|u|}{\sqrt{n}} \xrightarrow{n \rightarrow \infty} 0,$$

Since $\frac{|u|}{\sqrt{n}} \rightarrow 0$, there exists $N \in \mathbb{N}$ such that for all $n \geq N$ and all $k = 1, \dots, n$:

$$\left|\frac{u\sqrt{k}}{n}\right| \leq \frac{|u|}{\sqrt{n}} < \frac{\pi}{4},$$

so all arguments lie in $(-\frac{\pi}{4}, \frac{\pi}{4})$, where $\cos > 0$. Therefore, each factor $\cos(\frac{u\sqrt{k}}{n})$ is strictly positive for all $n \geq N$ and all $k = 1, \dots, n$. In particular $\varphi_{M_n}(u) > 0$, so we may take the logarithm of equation (10): for all $n \geq N$:

$$\ln \varphi_{M_n}(u) = \sum_{k=1}^n \ln \cos\left(\frac{u\sqrt{k}}{n}\right). \quad (11)$$

We now decompose each summand using standard Taylor expansions from analysis. The degree-2 Taylor expansion of cosine with Lagrange remainder gives:

$$\cos(x) = 1 - \frac{x^2}{2} + R_{\cos}(x), \quad |R_{\cos}(x)| \leq \frac{x^4}{24}, \quad (12)$$

and the degree-1 Taylor expansion of the logarithm gives:

$$\ln(1+y) = y + R_{\ln}(y), \quad |R_{\ln}(y)| \leq y^2 \quad \text{for all } |y| \leq \frac{1}{2}. \quad (13)$$

Setting $y_k := \cos(\frac{u\sqrt{k}}{n}) - 1$, we have from (12):

$$y_k := \cos\left(\frac{u\sqrt{k}}{n}\right) - 1 = 1 - \frac{u^2 k}{2n^2} + R_{\cos}\left(\frac{u\sqrt{k}}{n}\right) - 1 = -\frac{u^2 k}{2n^2} + R_{\cos}\left(\frac{u\sqrt{k}}{n}\right). \quad (14)$$

For the bound on $|y_k|$, we use the triangle inequality together with $|R_{\cos}(x)| \leq \frac{x^4}{24}$ and $k \leq n$:

$$|y_k| \leq \frac{u^2 k}{2n^2} + \left|R_{\cos}\left(\frac{u\sqrt{k}}{n}\right)\right| \leq \frac{u^2 k}{2n^2} + \frac{u^4 k^2}{24n^4} \leq \frac{u^2 n}{2n^2} + \frac{u^4 n^2}{24n^4} = \frac{u^2}{2n} + \frac{u^4}{24n^2}. \quad (15)$$

For large enough n , $|y_k| \leq \frac{1}{2}$ for all $k = 1, \dots, n$. Applying (13) with this y_k and summing over (11):

$$\begin{aligned} \ln \varphi_{M_n}(u) &= \sum_{k=1}^n [y_k + R_{\ln}(y_k)] \\ &\stackrel{(14)}{=} \sum_{k=1}^n \left[-\frac{u^2 k}{2n^2} + R_{\cos}\left(\frac{u\sqrt{k}}{n}\right) + R_{\ln}(y_k) \right] \\ &= -\frac{u^2}{2n^2} \sum_{k=1}^n k + \sum_{k=1}^n R_{\cos}\left(\frac{u\sqrt{k}}{n}\right) + \sum_{k=1}^n R_{\ln}(y_k). \end{aligned} \quad (16)$$

We evaluate the main term using $\sum_{k=1}^n k = \frac{n(n+1)}{2}$:

$$\frac{u^2}{2n^2} \sum_{k=1}^n k = \frac{u^2 n(n+1)}{4n^2} = \frac{u^2(n+1)}{4n} \xrightarrow{n \rightarrow \infty} \frac{u^2}{4}. \quad (17)$$

For the error terms, we use $\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6} \leq n^3$. First:

$$\left| \sum_{k=1}^n R_{\cos} \left(\frac{u\sqrt{k}}{n} \right) \right| \stackrel{(12)}{\leq} \left| \sum_{k=1}^n \frac{u^4 k^2}{24n^4} \right| = \frac{u^4}{24n^4} \sum_{k=1}^n k^2 \leq \frac{u^4 n^3}{24n^4} = \frac{u^4}{24n} \xrightarrow{n \rightarrow \infty} 0. \quad (18)$$

Second, from (15), we have $|y_k| \leq \frac{u^2}{2n} + \frac{u^4}{24n^2}$ and, as a bound, $\frac{u^2}{2n} + \frac{u^4}{24n^2} \leq \frac{u^2}{n}$ for all sufficiently large n , so that $y_k^2 \leq \frac{u^4}{n^2}$ for all $k = 1, \dots, n$, and therefore:

$$\left| \sum_{k=1}^n R_{\ln}(y_k) \right| \stackrel{(13)}{\leq} \sum_{k=1}^n y_k^2 \leq n \cdot \frac{u^4}{n^2} = \frac{u^4}{n} \xrightarrow{n \rightarrow \infty} 0. \quad (19)$$

Combining (17)–(19) with (16):

$$\ln \varphi_{M_n}(u) \xrightarrow{n \rightarrow \infty} -\frac{u^2}{4}, \quad \text{hence} \quad \varphi_{M_n}(u) \xrightarrow{n \rightarrow \infty} e^{-u^2/4}. \quad (20)$$

Step 4: Identification and application of Lévy's theorem. For $M \sim N(0, \frac{1}{2})$, Example 3.5.3 (5) gives:

$$\varphi_M(u) = e^{iu\mu - \frac{u^2\sigma^2}{2}} = e^{iu \cdot 0 - \frac{u^2 \cdot \frac{1}{2}}{2}} = e^{-u^2/4}.$$

The function $f(u) := e^{-u^2/4}$ is continuous at $u = 0$ with $f(0) = 1$, and we have shown $\varphi_{M_n}(u) \rightarrow f(u)$ for every $u \in \mathbb{R}$. By Lévy's Continuity Theorem (Theorem 4.2.4 b)), there exists a probability measure μ on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ such that the distributions of M_n converge weakly to μ and $\hat{\mu} = f$. Since $f = \varphi_M$ for $M \sim N(0, \frac{1}{2})$, the uniqueness of characteristic functions (Theorem 3.5.4) gives $\mu = P_M$ (i.e., the distribution of M), and therefore, by Definition 4.1.4 (ii):

$$M_n \xrightarrow{d} M \sim N\left(0, \frac{1}{2}\right). \quad (21)$$

No Convergence in Probability.

Suppose for contradiction that there exists a random variable Y on $(\Omega, \mathcal{F}, \mathbb{P})$ such that $M_n \xrightarrow{p} Y$. By Theorem 4.1.8a, this implies $M_n \xrightarrow{d} Y$. Since limits in distribution are unique (Theorem 3.5.4), this forces $Y \sim \mathcal{N}(0, \frac{1}{2})$ (by (21)), i.e. $M_n \xrightarrow{d} \mathcal{N}(0, \frac{1}{2})$. Since $2n \rightarrow \infty$ as $n \rightarrow \infty$, it follows that $M_{2n} \xrightarrow{p} Y$, and by the same argument:

$$M_{2n} \xrightarrow{d} \mathcal{N}\left(0, \frac{1}{2}\right). \quad (22)$$

Since both $M_n \xrightarrow{p} Y$ and $M_{2n} \xrightarrow{p} Y$, we can evaluate the difference $M_{2n} - M_n$. By the triangle inequality:

$$\begin{aligned} |M_{2n} - M_n| &\leq |M_{2n} - Y| + |Y - M_n| \\ \implies |M_{2n} - M_n| &\leq |M_{2n} - Y| + |M_n - Y| \end{aligned}$$

For any $\varepsilon > 0$, if we assume $|M_{2n} - M_n| > \varepsilon$, we have $|M_{2n} - Y| + |M_n - Y| \geq |M_{2n} - M_n| > \varepsilon$. This implies that at least one of the terms on the left-hand side must be greater than $\varepsilon/2$. In

set notation, $\{|M_{2n} - M_n| > \varepsilon\} \subset \{|M_{2n} - Y| > \varepsilon/2\} \cup \{|M_n - Y| > \varepsilon/2\}$ (i.e., the event $\{|M_{2n} - M_n| > \varepsilon\}$ algebraically implies that the distance from at least one of the variables to Y must exceed $\varepsilon/2$). By the subadditivity of probability measures (Lemma 1.1.10e), we have:

$$\mathbb{P}(|M_{2n} - M_n| > \varepsilon) \leq \mathbb{P}\left(|M_{2n} - Y| > \frac{\varepsilon}{2}\right) + \mathbb{P}\left(|M_n - Y| > \frac{\varepsilon}{2}\right).$$

By the definition of convergence in probability (Definition 4.1.1), both terms on the right-hand side converge to 0 as $n \rightarrow \infty$. Consequently:

$$\mathbb{P}(|M_{2n} - M_n| > \varepsilon) \xrightarrow{n \rightarrow \infty} 0.$$

Hence, $M_{2n} - M_n \xrightarrow{p} 0$. Since the probability distribution of a constant 0 is the Dirac measure δ_0 (Definition 1.2.9a) and, by Theorem 4.1.8a, we have:

$$M_{2n} - M_n \xrightarrow{d} \delta_0 \tag{23}$$

Contradiction. We now derive a contradiction by showing that in fact $M_{2n} - M_n \xrightarrow{d} \mathcal{N}(0, \frac{1}{2})$. Writing:

$$\begin{aligned} M_{2n} - M_n &= \frac{S_{2n}}{2n} - \frac{S_n}{n} = \frac{\left(\sum_{k=1}^{2n} X_k\right) - 2\left(\sum_{k=1}^n X_k\right)}{2n} \\ &= \frac{\left(\sum_{k=1}^n X_k + \sum_{k=n+1}^{2n} X_k\right) - 2\left(\sum_{k=1}^n X_k\right)}{2n} \\ &= \frac{\sum_{k=n+1}^{2n} X_k - \sum_{k=1}^n X_k}{2n} \\ &= \frac{\sum_{k=1}^n (-X_k) + \sum_{k=n+1}^{2n} X_k}{2n}. \end{aligned}$$

We observe that by the symmetric distribution (1), we have $X_k \stackrel{d}{=} -X_k$ for each k (both take values $\pm k^{1/2}$ with equal probability $1/2$). Consequently, the random vector $(-X_1, \dots, -X_n, X_{n+1}, \dots, X_{2n})$ has the same joint distribution as $(X_1, \dots, X_{2n})$ because the components are still mutually independent and, therefore, their joint density is simply the product of their marginal distributions (which are given by $\mathbb{P}_{X_k}$ for every $k = 1, \dots, 2n$ as, by symmetry, $\mathbb{P}_{-X_k} = \mathbb{P}_{X_k}$). It follows that:

$$M_{2n} - M_n = \frac{\sum_{k=1}^n (-X_k) + \sum_{k=n+1}^{2n} X_k}{2n} \stackrel{d}{=} \frac{\sum_{k=1}^{2n} X_k}{2n} = M_{2n}. \tag{24}$$

The above, combined with $M_{2n} \xrightarrow{d} \mathcal{N}(0, \frac{1}{2})$ (from (22)), implies $M_{2n} - M_n \xrightarrow{d} \mathcal{N}(0, \frac{1}{2})$ which **contradicts** $M_{2n} - M_n \xrightarrow{d} \delta_0$ (from (23)): the normal distribution $\mathcal{N}(0, \frac{1}{2})$ is an absolutely continuous distribution with strictly positive variance, whereas δ_0 is a degenerate distribution (Definition 1.2.9a) where all probability mass is concentrated at the single point 0. Because a sequence cannot converge in distribution to two distinct probability measures (Theorem 3.5.4), the assumption $M_n \xrightarrow{p} Y$ must be false. Therefore, $(M_n)_{n \in \mathbb{N}}$ does not converge in probability.

Part d)

We proceed by contradiction. Suppose $M_n \xrightarrow{d} M$ for some random variable M on $(\Omega, \mathcal{F}, \mathbb{P})$. By Lévy's Continuity Theorem (Theorem 4.2.4a)), this implies that the characteristic functions converge pointwise:

$$\varphi_{M_n}(u) \xrightarrow{n \rightarrow \infty} \varphi_M(u) \quad \forall u \in \mathbb{R}. \quad (25)$$

In particular, φ_M must be continuous (Theorem 3.5.2c)) with $\varphi_M(0) = 1$. We will show that in fact $\varphi_{M_n}(u) \rightarrow 0$ for all $u \neq 0$, which contradicts $\varphi_M(0) = 1$ and the continuity of φ_M at 0.

Step 1: Explicit form of φ_{M_n} . Since X_k is symmetric around 0 (taking values k^λ and $-k^\lambda$ with equal probability $\frac{1}{2}$), its characteristic function is strictly real-valued for all $k \in \mathbb{N}$. Using Definition 3.5.1 and Euler's formula, the imaginary components cancel out:

$$\begin{aligned} \varphi_{X_k}(t) &= \mathbb{E}[e^{itX_k}] = \frac{1}{2} e^{itk^\lambda} + \frac{1}{2} e^{-itk^\lambda} \\ &= \frac{1}{2} (\cos(tk^\lambda) + i \sin(tk^\lambda)) + \frac{1}{2} (\cos(tk^\lambda) - i \sin(tk^\lambda)) \\ &= \cos(tk^\lambda), \quad t \in \mathbb{R}. \end{aligned} \quad (26)$$

Since $X_1, \dots, X_n$ are independent and $M_n = \frac{1}{n} \sum_{k=1}^n X_k$, we compute (using Theorem 3.5.2b) and Lemma 3.5.6):

$$\varphi_{M_n}(u) = \prod_{k=1}^n \varphi_{X_k}\left(\frac{u}{n}\right) = \prod_{k=1}^n \cos\left(\frac{u k^\lambda}{n}\right), \quad u \in \mathbb{R}. \quad (27)$$

Since all factors in (27) are real, we have $|\varphi_{M_n}(u)| = \prod_{k=1}^n \left| \cos\left(\frac{u k^\lambda}{n}\right) \right|$.

Step 2: An upper bound on $|\varphi_{M_n}(u)|$. Fix $u \neq 0$. To bound the product in (27), we take the natural logarithm of its absolute value, converting the product into a sum:

$$\ln |\varphi_{M_n}(u)| = \sum_{k=1}^n \ln \left| \cos\left(\frac{u k^\lambda}{n}\right) \right|. \quad (28)$$

To bound this sum from above, we rely on the following inequality:

$$\ln \cos(x) \leq -\frac{x^2}{2} \quad \text{for all } |x| < \frac{\pi}{2}. \quad (29)$$

Proof of inequality: Define $g(x) := \ln \cos(x) + \frac{x^2}{2}$ on the interval $[0, \frac{\pi}{2})$. At $x = 0$, $g(0) = \ln(1) + 0 = 0$. The derivative is $g'(x) = -\tan(x) + x$. From standard Taylor bounds, $\tan(x) \geq x$ for all $x \in [0, \frac{\pi}{2})$, meaning $g'(x) \leq 0$. Since g starts at 0 and is non-increasing, $g(x) \leq 0$ on $[0, \frac{\pi}{2})$. Because both $\ln \cos(x)$ and $x^2/2$ are even functions (symmetric around 0), this bound extends to $(-\frac{\pi}{2}, 0]$. The restriction $|x| < \frac{\pi}{2}$ is strictly necessary because $\cos(x) \leq 0$ outside this interval, leaving the real-valued logarithm undefined.

To apply (29) to the terms in our sum (28), we must identify the indices k for which the argument $x = \frac{|u|k^\lambda}{n}$ satisfies the required domain constraint. We solve for k :

$$\frac{|u|k^\lambda}{n} < \frac{\pi}{2} \implies k^\lambda < \frac{n\pi}{2|u|} \implies k < \left(\frac{n\pi}{2|u|} \right)^{1/\lambda}.$$

To cleanly split the sum (28), we define an integer truncation limit m_n that captures exactly the indices satisfying this strict inequality, capped at the maximum index n :

$$m_n := \min \left\{ n, \left\lfloor \left(\frac{n\pi}{2|u|} \right)^{1/\lambda} \right\rfloor \right\}. \quad (30)$$

By this construction, we have split the summation in (28) into two parts:

1. **The Tail** ($k > m_n$): This occurs only if $m_n < n$. For these terms, $\frac{uk^\lambda}{n}$ equals or exceeds $\frac{\pi}{2}$, so we use the trivial bound $\left| \cos\left(\frac{uk^\lambda}{n}\right) \right| \leq 1$. The logarithm of any value in $(0, 1]$ is ≤ 0 . Because these terms contribute ≤ 0 to the sum, dropping them preserves the upper bound.
2. **The Core** ($k \leq m_n$): For these terms, $\frac{|u|k^\lambda}{n}$ is strictly inside $(0, \frac{\pi}{2})$. The cosine is strictly positive (so the absolute value bars can be dropped) and the domain requirement for (29) is satisfied.

Dropping the tail terms in (28) gives an upper bound consisting only of the core terms (as argued in point 1, dropping them preserves the upper bound). Because the cosine is an even function, $\cos(\frac{uk^\lambda}{n}) = \cos(\frac{|u|k^\lambda}{n})$. Furthermore, because this argument is strictly in $(0, \frac{\pi}{2})$ for these core terms, the cosine is strictly positive, allowing us to drop the outer absolute value bars:

$$\ln |\varphi_{M_n}(u)| \leq \sum_{k=1}^{m_n} \ln \cos\left(\frac{|u|k^\lambda}{n}\right). \quad (31)$$

We apply the established inequality (29) directly to the remaining terms:

$$\ln |\varphi_{M_n}(u)| \leq \sum_{k=1}^{m_n} -\frac{1}{2} \left(\frac{|u|k^\lambda}{n} \right)^2 = -\frac{u^2}{2n^2} \sum_{k=1}^{m_n} k^{2\lambda}. \quad (32)$$

Step 3: Showing the right-hand side of (32) diverges to $-\infty$. We want to show that the negative term in our upper bound grows unboundedly in magnitude as $n \rightarrow \infty$. We first establish a continuous lower bound for the discrete sum $\sum_{k=1}^{m_n} k^{2\lambda}$. Because $\lambda > 0$, the function $f(x) = x^{2\lambda}$ is strictly increasing on $[0, \infty)$. Consequently, a right-endpoint Riemann sum with step size $\Delta x = 1$ strictly overestimates the area under the curve. We can therefore lower-bound the right-endpoint Riemann sum with the corresponding definite integral:

$$\sum_{k=1}^{m_n} k^{2\lambda} \geq \int_0^{m_n} x^{2\lambda} dx = \frac{m_n^{2\lambda+1}}{2\lambda+1}. \quad (33)$$

Substituting this continuous bound into our previously established upper bound (32), we get:

$$\ln |\varphi_{M_n}(u)| \leq -\frac{u^2}{2n^2} \sum_{k=1}^{m_n} k^{2\lambda} \leq -\frac{u^2}{2(2\lambda+1)} \frac{m_n^{2\lambda+1}}{n^2}. \quad (34)$$

To evaluate the limit of this bound, we must consider how the truncation limit $m_n = \min \left\{ n, \left\lfloor \left(\frac{n\pi}{2|u|} \right)^{1/\lambda} \right\rfloor \right\}$ behaves for large n . The active constraint in the minimum function depends only on the value of λ (u only affects the scaling). We split this into two cases:

- **Case 1:** $1/2 < \lambda \leq 1$. Here, the exponent $1/\lambda \geq 1$. This means the term inside the floor function grows linearly or super-linearly with n . For sufficiently large n , this term will strictly exceed n . Therefore, the minimum resolves to the first argument: $m_n = n$. Substituting $m_n = n$ into the right-hand side of (34) yields:

$$-\frac{u^2}{2(2\lambda+1)} \frac{n^{2\lambda+1}}{n^2} = -\frac{u^2}{2(2\lambda+1)} n^{2\lambda-1}.$$

Because $\lambda > 1/2$, the exponent $(2\lambda - 1)$ is strictly positive. As $n \rightarrow \infty$, the term $n^{2\lambda-1} \rightarrow \infty$, and, thus, the entire bound diverges to $-\infty$.

- **Case 2:** $\lambda > 1$. Here, the exponent $1/\lambda < 1$. The term inside the floor function grows sub-linearly with n . For sufficiently large n , it will be strictly less than n . Therefore, the minimum resolves to the second argument: $m_n = \left\lfloor \left(\frac{n\pi}{2|u|} \right)^{1/\lambda} \right\rfloor$. To handle the floor function algebraically, we use the property that $\lfloor x \rfloor \geq \frac{x}{2}$ for all $x \geq 2$. Since our term grows to infinity, this holds true for all large n . Thus:

$$m_n \geq \frac{1}{2} \left(\frac{n\pi}{2|u|} \right)^{1/\lambda}.$$

Substituting this lower bound for m_n into the right-hand side of (34) yields:

$$\begin{aligned} -\frac{u^2}{2(2\lambda+1)n^2} \left[\frac{1}{2} \left(\frac{n\pi}{2|u|} \right)^{1/\lambda} \right]^{2\lambda+1} &= -\underbrace{\left[\frac{u^2}{2(2\lambda+1)2^{2\lambda+1}} \left(\frac{\pi}{2|u|} \right)^{\frac{2\lambda+1}{\lambda}} \right]}_{:=C_{\lambda,u}} \cdot \frac{n^{\frac{2\lambda+1}{\lambda}}}{n^2} \\ &= -C_{\lambda,u} \cdot n^{(2+1/\lambda)-2} \\ &= -C_{\lambda,u} \cdot n^{1/\lambda}. \end{aligned}$$

Here, $C_{\lambda,u} > 0$ is a strictly positive constant depending only on λ and u . Because $\lambda > 0$, the exponent $1/\lambda$ is strictly positive. As $n \rightarrow \infty$, the term $n^{1/\lambda} \rightarrow \infty$, again causing the entire bound to diverge to $-\infty$.

In both cases, for any $\lambda > 1/2$, the upper bound of $\ln |\varphi_{M_n}(u)|$ strictly diverges to $-\infty$. Consequently:

$$\ln |\varphi_{M_n}(u)| \xrightarrow{n \rightarrow \infty} -\infty \implies |\varphi_{M_n}(u)| \xrightarrow{n \rightarrow \infty} 0 \quad \text{for all } u \neq 0. \quad (35)$$

Step 4: Contradiction. If $M_n \xrightarrow{d} M$, then from (25) and (35) it follows that $\varphi_M(u) = 0$ for all $u \neq 0$. However, every characteristic function satisfies $\varphi_M(0) = 1$ (Theorem 3.5.2 a)) and is continuous everywhere (Theorem 3.5.2 c)). A function that vanishes identically for all $u \neq 0$ but equals 1 at $u = 0$ cannot be continuous at 0. This is a contradiction. Hence $(M_n)_{n \in \mathbb{N}}$ does not converge in distribution for any $\lambda > \frac{1}{2}$.

Since convergence in probability implies convergence in distribution (Theorem 4.1.8 a)), and $\mathbb{P}$ -almost sure convergence implies convergence in probability (Theorem 4.1.2 a)), it follows that $(M_n)_{n \in \mathbb{N}}$ also does not converge in probability or $\mathbb{P}$ -almost surely. □